

UGQ301 Project Report

Predicting Electron Interactions as an Evolving Graph

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GitHub Repository:

<https://github.com/Galabavamsi/Electron-GNN>

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Submission Date: May 7, 2026

Department / Course: UGQ301

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Abstract

This project studies two complementary machine-learning routes for accelerating quantum dynamics from real RT-TDDFT simulation data. The first route, **Electron-GNN**, predicts sparse absorption-spectrum peak parameters directly from molecular geometry and reconstructs the observable spectrum analytically. The second route, **Temporal-GAT**, predicts the next electron-density field over a dense 3D grid using a graph built on spatial and temporal relationships between grid points. Together, these pipelines address two different but connected scientific targets mentioned in the presentation: **dipole moment / spectral prediction** and **probability density prediction**.

The spectral branch shows that a physics-guided two-tower hybrid model can outperform either tower alone, reaching an average spectral overlap of **0.570**, which is **16.9% better than the V1 baseline** and **6.0% better than V2 alone**. The dense-density branch shows that graph-based forecasting can reconstruct the absolute electron density of ammonia with $R^2 = 1.000000$ on active nodes, **100% of active nodes within 0.5% error**, and an electron conservation error of only **0.00006%**. The main conclusion across the whole project is that the architectures are viable and physically meaningful, but both branches are presently constrained by data scale.

1 Introduction

Real-Time Time-Dependent Density Functional Theory (RT-TDDFT) is a powerful tool for studying how electrons respond to external perturbations, but it is also computationally expensive. This makes long simulations, high-frequency resolution, and molecule-scale screening difficult. The central idea of this UGQ301 project is to replace part of that expensive numerical workload with graph neural networks that operate on physically structured data.

The overall project evolved into two independent but related goals:

Goal 1: Electron-GNN predicts the sparse spectral peak set (ω_k, B_k) from molecular geometry, then reconstructs the absorption spectrum using Lorentzian broadening.

Goal 2: TemporalGAT predicts the next electron-density field from a short history of previous density fields over a large spatial grid.

This dual-goal structure is important. The first path compresses the physics into a small set of observable parameters. The second keeps the full spatial field and tries to learn the time evolution directly. Together they provide a broad view of how graph learning can support quantum dynamics.

2 Project Motivation and Data Context

The project presentation highlights three shared difficulties across the dataset and modeling pipeline:

Table 2.1: Shared project challenges

Challenge		Why it matters
Extreme range	dynamic	Electron density values span roughly 17 orders of magnitude, making naive learning unstable.
Tiny changes	temporal	Consecutive density updates are very small relative to the absolute field, so a model can appear accurate while learning little true dynamics.
Small dataset size		Both branches work with very limited chemistry coverage, which prevents strong generalization claims.

The data source for both goals is RT-TDDFT simulation output. In the spectral branch, the raw dipole trace is transformed into sparse transition peaks. In the dense branch, the raw time series of electron density values across a 3D grid is treated as a large evolving graph problem.

Before arriving at the final project framing, the team also considered standard MLPs on flattened grids, CNNs on volumetric grids, RNN/LSTM sequence models, and classical signal-processing pipelines without machine learning. The final report focuses on the two GNN-based pipelines because they best preserve physical structure while remaining computationally practical.

3 Goal 1: Spectral Peak and Dipole Moment Prediction

3.1 Objective

The goal of the Electron-GNN branch is to predict an absorption spectrum directly from molecular structure without running a fresh RT-TDDFT simulation. Instead of regressing the full spectrum point by point, the model predicts a sparse set of transition frequencies and amplitudes:

$$S(\omega) = \sum_k \frac{B_k \gamma}{(\omega - \omega_k)^2 + \gamma^2}. \quad (3.1)$$

This is physically meaningful and data efficient because the spectrum is determined by a small number of peaks rather than a dense high-resolution curve.

3.2 End-to-End Pipeline

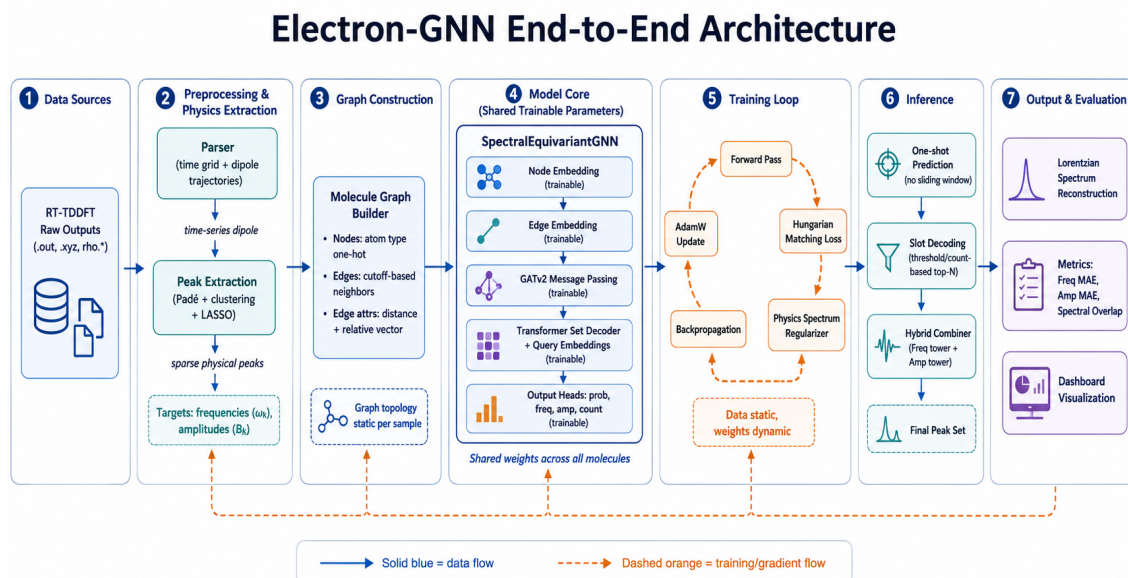


Figure 3.1: PPT architecture for Goal 1, showing the complete Electron-GNN workflow from RT-TDDFT data through inference and evaluation.

The pipeline proceeds in six stages:

1. Parse RT-TDDFT outputs from ReSpect files.
2. Extract physical peaks from the dipole trace using a physics-guided signal-processing workflow.
3. Convert molecular geometry into a PyTorch Geometric graph.
4. Train two specialized graph models for complementary subproblems.
5. Combine both towers through hybrid inference.
6. Reconstruct the final spectrum and compare it to the extracted ground truth.

3.3 Physics Extraction

The presentation refers to this as the **Hauge pipeline**. The dipole signal is filtered and compressed into a sparse peak representation using:

1. Butterworth low-pass filtering.
2. Fourier-Pade pole extraction.
3. K-Means cleanup of numerical ghost poles.
4. Positive LASSO regression for nonnegative amplitudes.

This step produces the supervised targets used by the model. On the current dataset, the extracted labels are:

Table 3.1: Extracted peak counts for Goal 1

Molecule	True peaks
Ammonia (NH_3)	39
Water (H_2O)	55
Total	94

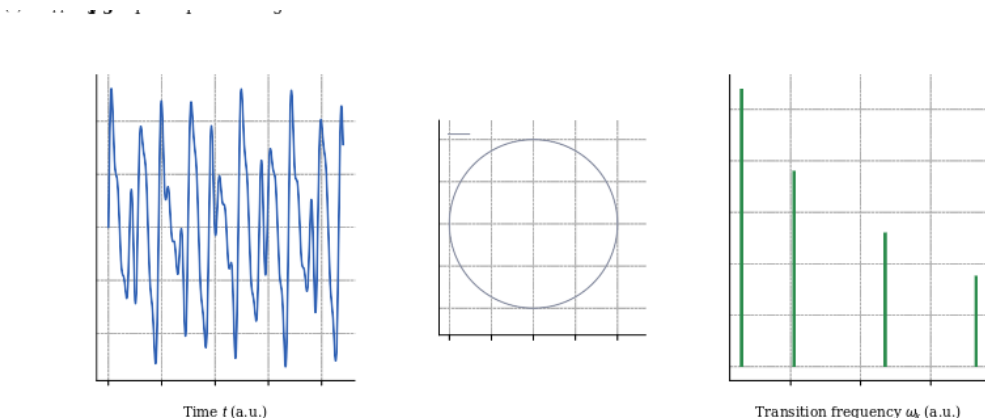


Figure 3.2: Physics-guided peak extraction used to convert dipole traces into sparse transition parameters.

3.4 Graph Representation and Model Design

Each molecule is represented as an atomic graph with one-hot node features over $[\text{H}, \text{C}, \text{N}, \text{O}, \text{F}]$ and edge features $[\text{distance}, \text{dx}, \text{dy}, \text{dz}]$ for atom pairs inside a cutoff radius of 5.0 a.u.

The final production design is a **two-tower system**:

- **V1 frequency tower:** a simpler architecture that is strong at placing peak frequencies.
- **V2 amplitude tower:** a GATv2 encoder plus Transformer-style set decoder that is better at amplitudes and peak count prediction.
- **V3 hybrid inference:** a matching-based merger that uses V1 for primary frequencies and V2 for amplitudes and cardinality.

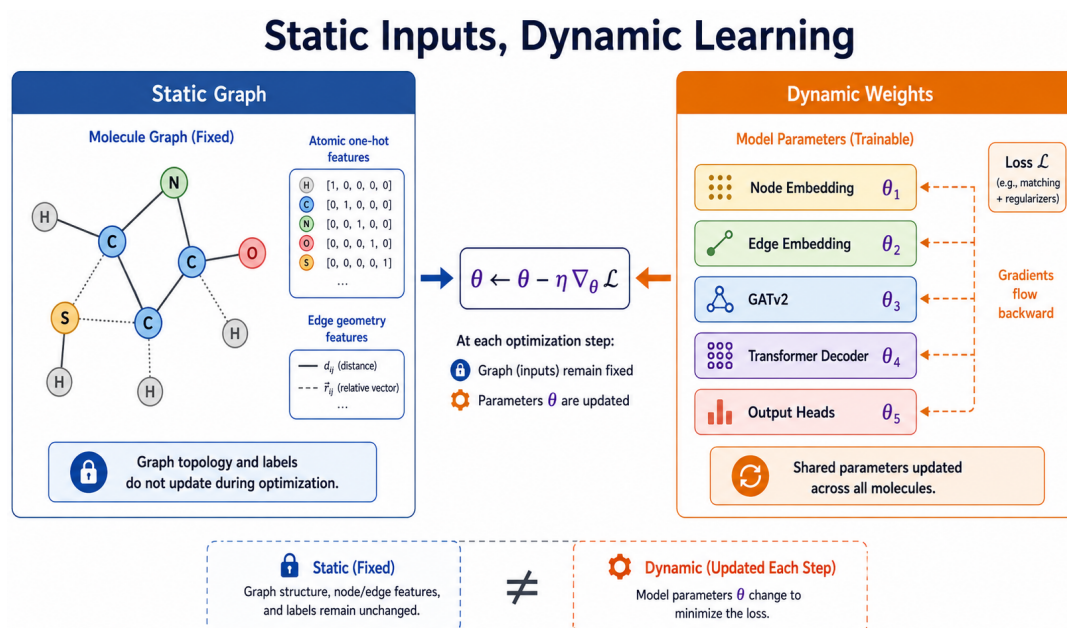


Figure 3.3: PPT architecture view of static molecular-graph inputs and dynamically updated shared trainable parameters in Electron-GNN.

This design exists because the dataset is tiny. On only two molecules, a single model does not learn frequency precision and amplitude calibration equally well. Splitting the task gives better practical behavior.

3.5 Training Objective

The spectral branch uses a physics-informed objective with two major parts:

1. **Hungarian bipartite matching loss:** this resolves the permutation ambiguity of unordered peak sets and penalizes frequency error, amplitude error, unmatched slots, and count mismatch.
2. **Auto-differential spectrum regularizer:** this reconstructs the time-domain signal and spectral curve inside PyTorch so that predictions are judged not only as lists of peaks, but also as observable physics.

The baseline V2 training run reduced bipartite loss by **91.9%** and spectrum loss by **79.7%**, with the best checkpoint saved at epoch 111.

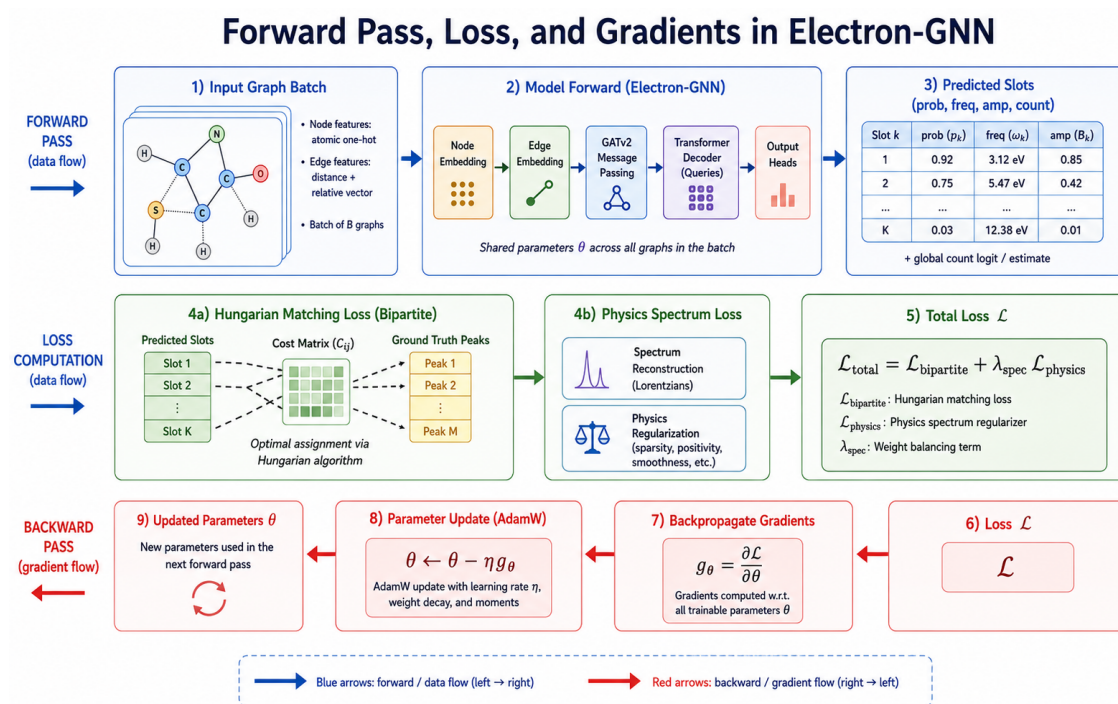


Figure 3.4: PPT training-and-loss architecture for Goal 1, showing forward pass, Hungarian matching, physics spectrum loss, and gradient flow.

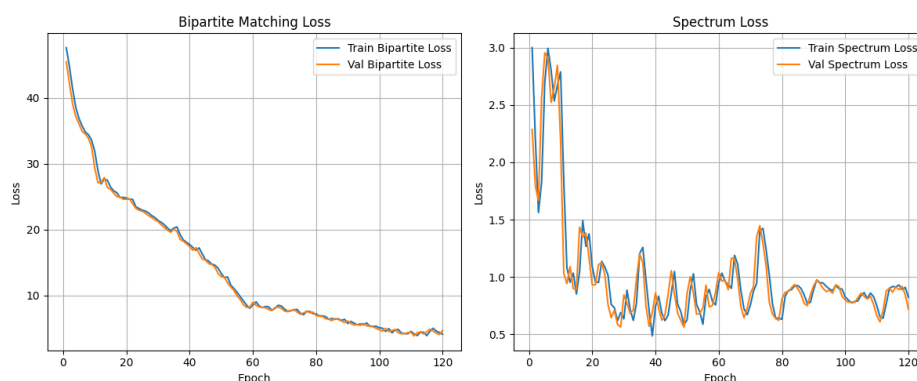


Figure 3.5: V2 training curves showing stable optimization and strong loss reduction on the baseline checkpoint.

3.6 Main Results

The key quantitative message from the PPT is that each tower is best at a different subtask, and the hybrid gives the best overall spectral match.

Table 3.2: Average performance for Goal 1

Model	Avg Freq MAE	Avg Amp MAE	Avg Overlap
V1	0.0373	2.13×10^{-3}	0.488
V2	0.5196	8.67×10^{-5}	0.537
Hybrid V3	0.1310	1.18×10^{-4}	0.570

Interpretation:

- V1 is the best frequency prior.
- V2 is the best amplitude learner.
- Hybrid V3 gives the best overall spectral overlap.
- The hybrid result is **16.9% better than V1 alone** and **6.0% better than V2 alone** on the overlap metric used in the presentation.

Per-molecule diagnostics also support the quantitative results.

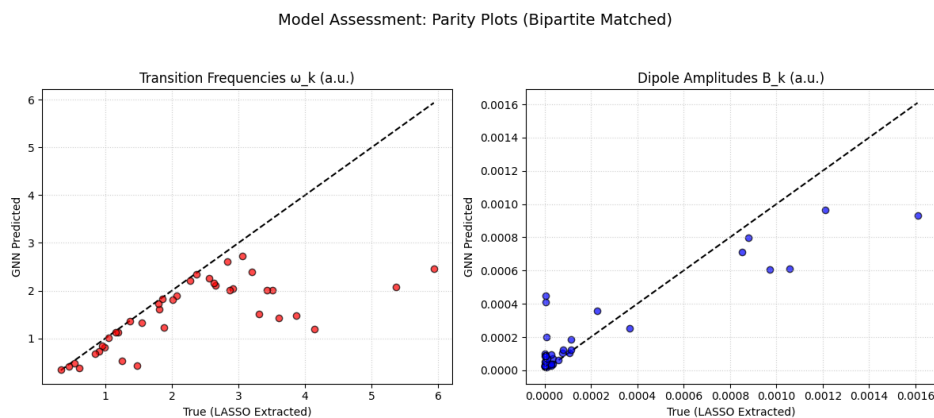


Figure 3.6: Ammonia parity plots comparing predicted and true frequencies and amplitudes after Hungarian matching.

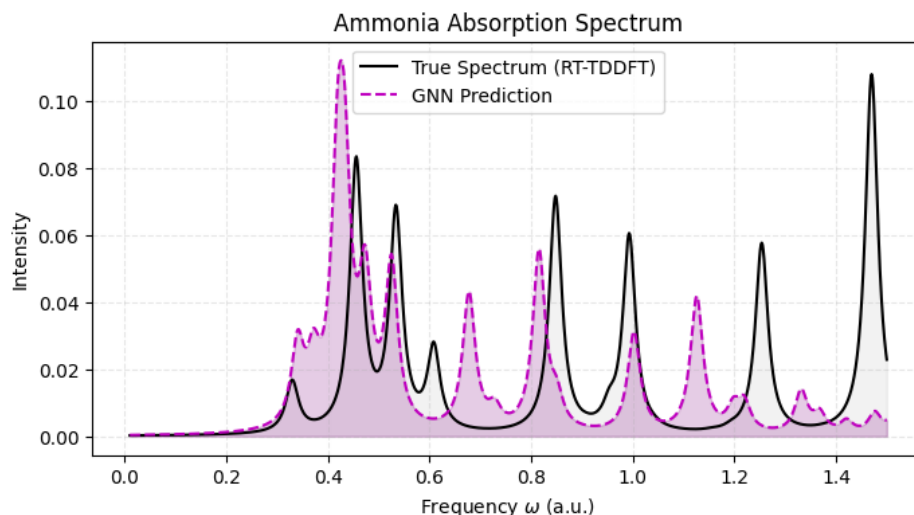


Figure 3.7: Reconstructed ammonia spectrum from the predicted peak set versus the extracted ground-truth spectrum.

3.7 What This Goal Demonstrates

The Electron-GNN branch demonstrates that a full end-to-end workflow is now in place:

RT-TDDFT output $\rightarrow$ *physics extraction* $\rightarrow$ *molecular graph* $\rightarrow$ *GNN prediction* $\rightarrow$
hybrid inference $\rightarrow$ *spectrum reconstruction*

That is a strong outcome because it converts a long simulation-and-postprocessing workflow into a compact inference problem. At the same time, the presentation is careful about the limitation: the model is operational, but the chemistry dataset is too small to claim broad generalization.

4 Goal 2: Probability Density Prediction with TemporalGAT

4.1 Objective

The second branch predicts the next electron-density field of ammonia from the previous five timesteps. Here the target is not a sparse spectrum but a dense 3D field over **10,540 spatial grid points**. This is the branch described in the notebook materials as **probability density prediction**.

The learning task can be summarized as:

$$f(\rho_{t-4}, \rho_{t-3}, \rho_{t-2}, \rho_{t-1}, \rho_t) \rightarrow \rho_{t+1}. \quad (4.1)$$

4.2 Data and Preprocessing

The notebook branch uses:

Table 4.1: Dataset summary for Goal 2

Property	Value
Molecule	Ammonia (NH ₃)
Timesteps	401
Grid points	10,540
Training samples after sliding windows	396
Train / Val / Test split	316 / 39 / 41

The preprocessing pipeline addresses the hardest numerical issues in the dataset:

1. Apply `log1p` to compress the raw density range.
2. Predict temporal deltas instead of raw absolute density.
3. Standardize the deltas to unit scale.
4. Use soft node weights so static vacuum-like regions do not dominate training.

The notebook reports that only **60.8%** of nodes are truly active, while **39.2%** are nearly static. This makes the problem deceptively hard: a model can look good on absolute density while learning very little about the actual temporal change.



Figure 4.1: Dataset diagnostics for the density-prediction branch, including skewness, delta scale, and active-versus-static node behavior.

4.3 Graph Construction and Model

Instead of a molecular graph over atoms, this branch builds a graph over density grid points using two complementary edge sets:

Table 4.2: Edge construction for Goal 2

Edge type	Method	k
Spatial edges	Euclidean kNN in 3D space	10
Feature edges	Cosine kNN on temporal signatures	8

After merging duplicate connections, the graph contains about **184,018 edges** with an average node degree of about **17.5**.

The forecasting model is a **TemporalGAT** with about **79,489 trainable parameters**. It uses:

- input projection from a 5-step temporal feature vector,
- four GAT blocks with residual connections and normalization,
- an MLP output head for per-node predictions.

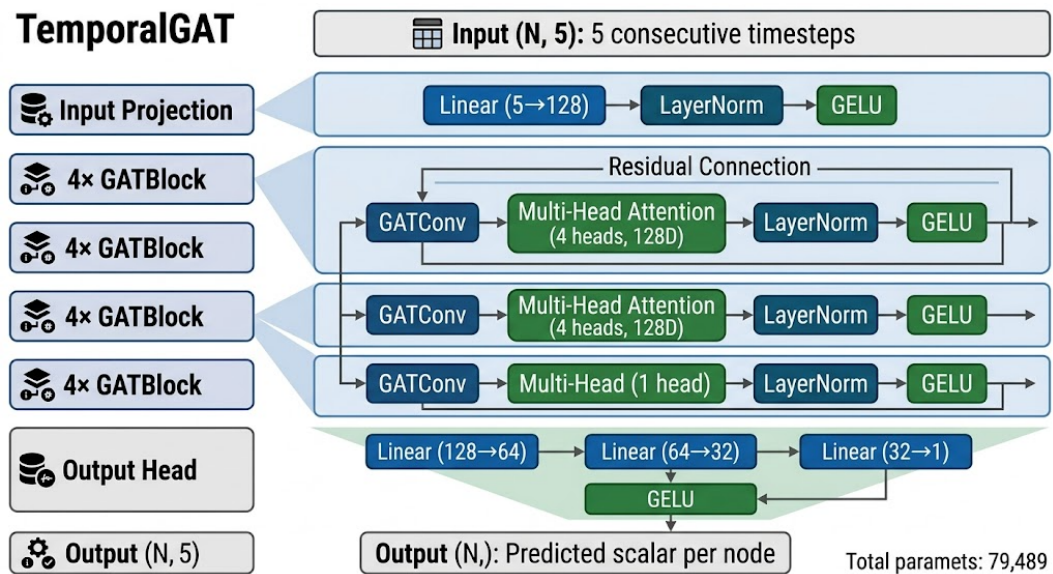


Figure 4.2: PPT architecture for Goal 2, showing the TemporalGAT network used for probability density prediction.

The training objective is also physics aware:

$$L_{\text{total}} = L_{\text{weighted mse}} + 0.20L_{\text{conservation}} + 0.02L_{\text{smoothness}}. \quad (4.2)$$

This ensures the model is judged not only on pointwise accuracy, but also on total electron conservation and spatial coherence.

4.4 Training Behavior

Training was stable and modestly underfit rather than overfit, which is a reasonable outcome for a deliberately small model on a limited dataset. The notebook reports:

Table 4.3: Training summary for Goal 2

Metric	Value
Training time	541 s
Best epoch	88
Best validation loss	2.645541



Figure 4.3: Training curves for the TemporalGAT model, including the train-validation loss ratio used to monitor overfitting.

4.5 Main Results

The central result of the notebook branch is a scientific paradox that the presentation explicitly discusses: the model is excellent at reconstructing the absolute density field, but weak at predicting the small timestep-to-timestep delta itself.

Table 4.4: Main quantitative results for Goal 2

Metric space	Result
Scaled delta R^2	0.000413
Raw delta R^2 on active nodes	0.000420
Absolute density R^2 on active nodes	1.000000
Within 0.5% on active nodes	100.00%
Electron conservation error	0.00006%
NRMSE on active nodes	0.00025%

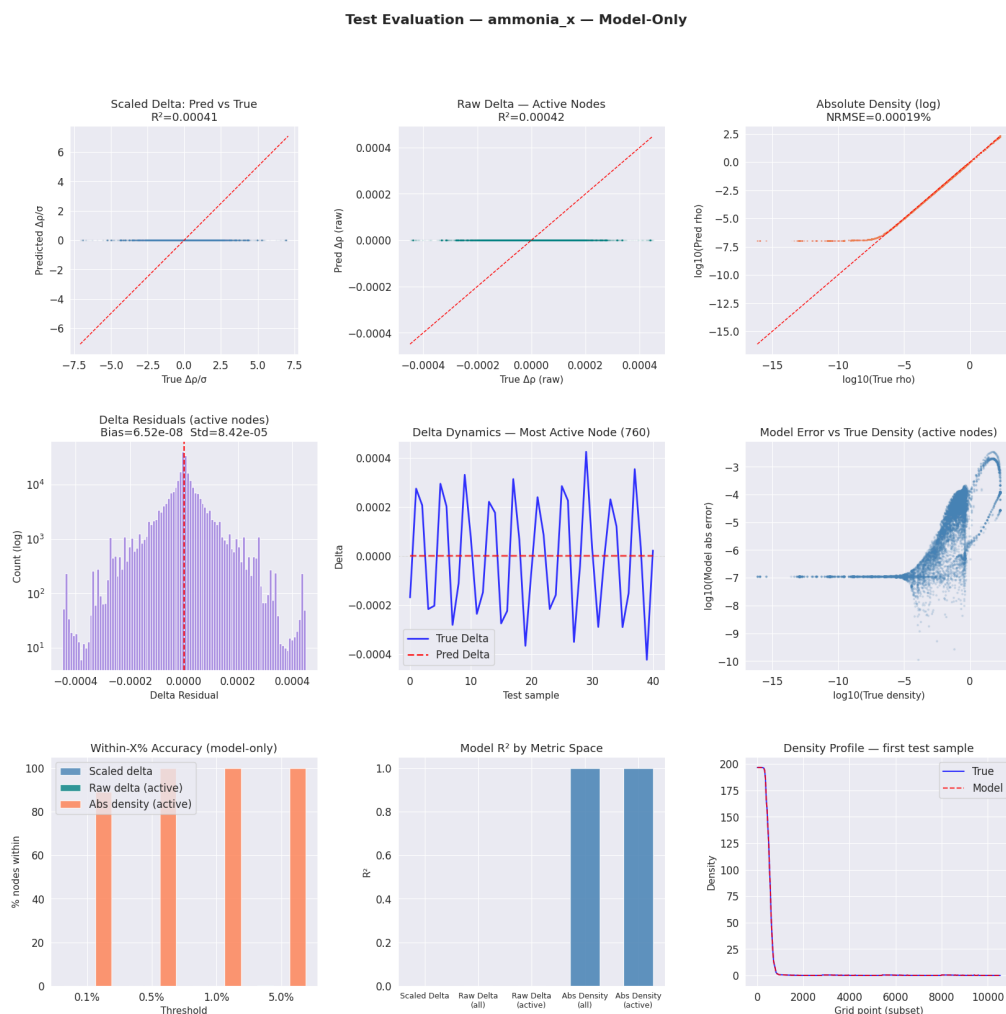


Figure 4.4: Test evaluation diagnostics showing near-zero explanatory power on the delta target but near-perfect reconstruction in absolute-density space.

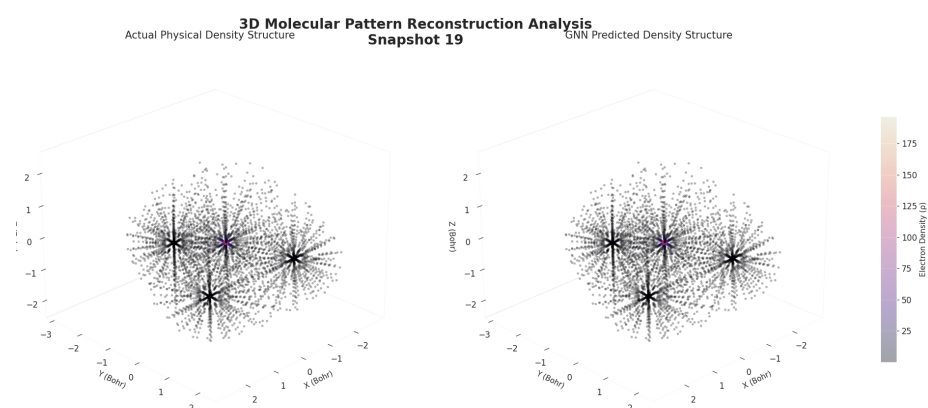


Figure 4.5: 3D reconstruction confirming that the predicted density field preserves the spatial structure of the ammonia molecule with very low visible distortion.

This means the model has learned a very strong near-identity reconstruction prior, but not yet the fine temporal dynamics that would make it a true learned propagator. That is

still a meaningful result: it shows that physically constrained graph models can preserve structure and total charge extremely well, even when the actual next-step signal is tiny.

5 Comparative Discussion

The two branches solve different scientific subproblems and should not be treated as direct competitors. Instead, they are complementary views of the same research direction.

Table 5.1: Comparison between the two project goals

Aspect	Goal 1: Electron-GNN	Goal 2: TemporalGAT
Prediction target	Sparse spectral peaks	Dense next-step electron density
Input	Molecular geometry	Five historical density snapshots
Graph nodes	Atoms	Spatial grid points
Key inductive bias	Set prediction with hybrid specialization	Spatiotemporal message passing with physics constraints
Best result	Avg overlap = 0.570	Abs. density $R^2 = \mathbf{1.000000}$
Main weakness	Too little chemistry diversity	Delta-level dynamics remain weak
Scientific takeaway	Sparse observable prediction is feasible	Full-field reconstruction is feasible

The common lesson is that **physics-aware graph learning works**, but what it achieves depends strongly on how the problem is represented.

- The spectral branch succeeds by reducing the problem to sparse physically meaningful quantities.
- The density branch succeeds by preserving geometry and conservation laws over a large graph.
- Both branches reveal that model design is no longer the only bottleneck; dataset scale and evaluation regime now matter more.

6 Conclusion

This UGQ301 project successfully produced two complete graph-based pipelines for ML-accelerated quantum dynamics. The first pipeline, Electron-GNN, maps molecular geometry to absorption spectra through sparse peak prediction and hybrid inference. The second pipeline, TemporalGAT, forecasts dense electron density fields with excellent structural fidelity and extremely strong charge conservation.

The strongest project-wide conclusion matches the presentation: **the architectures work, the physics constraints help, and data is now the main bottleneck**. The spectral branch already shows a practical hybrid advantage in overlap-based evaluation, while the density branch shows that full-field reconstruction can be learned with remarkable accuracy. Together, these results make a strong foundation for the next stage of the project: scaling the dataset, strengthening generalization tests, and turning these proof-of-concept pipelines into robust scientific tools.